P525/1
CHEMISTRY
Paper 1
(Theory)
Jul./Aug. 2026
$2\frac{3}{4}$ hours

WAKISO-KAMPALA TEACHERS' ASSOCIATION (WAKATA)

WAKATA MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

CHEMISTRY

(Theory)

KATA 2026

Paper 1

2 hours 45 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two** *sections;* **A** *and* **B**. *It has* **six** *examination items*

Section **A** *has* **two compulsory** *items.*

Section **B** *has* **two** *parts.* **I** *and* **II.** *Respond to only one item from each part*

Respond to **four** *items in all.*

Responses of both sections must be written in the answer booklet provided.

Responses to each item must start on a **fresh page.**

All items carry equal scores

Any additional item(s) responded to will not be scored.

Where necessary, use the following, **C=12, H=1, O=16, Cl=35.5,** *1 mole of a gas at room temperature occupies* $24dm^3$

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SECTION A

Respond to **all** *items in this section.*

Item 1.

Calcium chloride ($CaCl_2$) is an important industrial chemical widely used in road construction to melt ice and snow because of its ability to absorb moisture and lower the freezing point of water.

A road construction company uses calcium chloride to melt ice and snow on roads during cold weather. However, the road construction engineer wants to determine the lattice energy and enthalpy of solution of calcium chloride, as well as understand the effect of temperature on its solubility in water, in order to assess its suitability for use. She also wants to determine whether the formation of calcium chloride in an electrochemical cell provides evidence that the reaction between calcium and chlorine is thermodynamically feasible.

She carried out research and obtained the following information, but she could not use it effectively:

$Ca^{2+}(aq) + 2e^- \rightarrow Ca(s)$, $E° = -2.87$ V

$Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$, $E° = +1.36$ V

Atomization energy of calcium	$Ca(s) \rightarrow Ca(g)$	+193 kJ/mol
First ionization energy of calcium	$Ca(g) \rightarrow Ca^+(g)$	+ 590 kJ/mol
Second ionization energy of calcium	$Ca^+(g) \rightarrow Ca^{2+}(g)$	+ 1150 kJ
Atomization energy of Cl	$\frac{1}{2} Cl_2(g) \rightarrow Cl(g)$	+ 122 kJ/mol
First electron affinity of Cl	$Cl(g) \rightarrow Cl^-(g)$	-349 kJ/mol
Enthalpy of formation of $CaCl_2$		-795 kJ/mol
Hydration enthalpy of calcium ions		-1650 kJ/mol
Hydration enthalpy of chloride ions		-381 kJ/mol

She has been referred to you for assistance.

Task:

As a chemistry student, help the construction engineer address her concerns.

Item 2.

Food additives play an important role in modern food production and preservation. Some of these additives are naturally derived from fruits.

A food manufacturing company is developing a new artificial pineapple flavour for use in frozen desserts. The flavouring compound is the ester ethyl butanoate, which has a characteristic pineapple aroma. To ensure that the product meets quality standards, the

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company's chemists must first synthesise the ester through esterification and then determine its purity using freezing-point measurements.

A chemist reacts butanoic acid with ethanol in the presence of concentrated sulphuric acid to produce ethyl butanoate. After purification, a sample of the ester is tested for its physical properties and purity. A small amount of impurity remains in the product, affecting its freezing point.

The pure ester has a freezing point of −93.0°C, whereas the synthesised sample freezes at −95.5°C. The cryoscopic constant (Kf) for ethyl butanoate is 7.2 °C kg mol^{-1}. A sample containing 0.50 g of impurity is dissolved in 100 g of ethyl butanoate.

However, the chemist wants to know how to use the above information to evaluate the suitability of the product. You have been contacted to assist with the analysis.

Task:
As a chemistry student, prepare a comprehensive report to assist the company in evaluating the suitability of the product.

SECTION B

PART I

Respond to only **one** *item from this part*

Item 3.

Differences in melting points allow engineers to select materials that can either withstand high temperatures or melt easily when required. Similarly, bond energies determine how readily substances react during chemical manufacturing processes.

A company that manufactures electronic circuit boards used in computers and mobile phones uses components made from Group IV and Group VII elements. To guide the selection of suitable materials, the research team collected data on the melting points of Group IV elements and the bond energies of Group VII elements, as shown below.

Group (IV) elements	**Melting points (^{0}C)**	**Group (VII) elements**	**Bond energy (KJ/mol)**
Carbon	3730	Fluorine	158
Silicon	1410	Chlorine	242
Germanium	937	Bromine	193
Tin	232	Iodine	151
Lead	327		

The research team wants an analysis of the trends in the physical properties of the elements shown above. They also want to understand the reactions of Group IV elements with air and the reactions of Group VII elements with water.
The company manager is interested in evaluating the applications of these elements and the environmental risks associated with their use. You have been approached for assistance.

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Task:
As a chemistry student, prepare a comprehensive report addressing the above concerns.

Item 4.

Many construction materials are based on the chemical behaviour of Group II carbonates and Period 3 chlorides. Group II carbonates are widely found in limestone, which is extensively used in cement production, while Period 3 chlorides are important in water purification.

A chemical company produces Period 3 chlorides for use in water treatment. The factory uses a furnace heated by the decomposition of Group II carbonates and needs to determine which carbonate source and chloride product are most suitable for high-temperature applications.

The thermal stability of Group II carbonates and the melting-point behaviour of Period 3 chlorides are key chemical properties that determine their suitability for these industrial applications.
The company manager is also interested in understanding how the carbonates neutralise acidic effluents containing hydrochloric acid while safeguarding the environment from the risks associated with the company's operations.

Carbonates	$BeCO_3$	$MgCO_3$	$CaCO_3$	$SrCO_3$	$BaCO_3$
Decomposition temperature (°C)	100	350	900	1290	1350

Chlorides	NaCl	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_5	S_2Cl_2	Cl_2
Melting Point °C	808	714	192	-68	160	-76	-101

You are part of the team assigned to analyse the information above.

Task:
As a chemistry student, prepare a comprehensive report addressing the above concerns.

PART II

Respond to only **one** *item from this part.*

Item 5.

Alcohols are widely used as fuels and in consumer products, while alkyl halides are important in refrigeration systems and as industrial solvents.
A chemical company is investigating suitable methods for producing phenol and 2-methylpropan-2-ol for use in the manufacture of resins and antiseptics.

The company intends to prepare phenol from bromobenzene and 2-methylpropan-2-ol from propan-2-ol. It also intends to study the elimination reaction of propan-2-ol using concentrated sulphuric acid and the reaction between phenol and propanoyl chloride, including the reaction mechanisms involved.

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The company further wishes to evaluate the environmental and ethical implications of producing and using these chemicals.
The research team also wants to determine the physical properties of bromobenzene and propan-2-ol to assess their suitability for industrial applications.
They have consulted you for assistance.

Task:
As a chemistry student, help the team address the above challenges.

Item 6.

Customers prefer products that not only remove nail polish effectively but also leave a pleasant fragrance. Carbonyl compounds are widely used as fragrance ingredients because of their characteristic aromas.

A cosmetics company manufactures a scented nail-polish remover. Two important chemicals used in the production process are propanone (acetone) and benzaldehyde.
To ensure a steady supply of propanone, the company intends to prepare it from propan-1-ol, while benzene is to be converted into benzaldehyde.

Before using these chemicals in production, quality-control chemists examine their physical properties. They also study the nucleophilic addition reaction of hydrogen cyanide with propanone and the condensation reaction of benzaldehyde with 2,4-dinitrophenylhydrazine, including the mechanisms involved.

The company also intends to evaluate the environmental impact of these chemical products.
The company team wants to examine the chemicals and assess their suitability for the intended industrial applications. You have been contacted to assist the team.

Task:
As a chemistry student, help the team address their concerns and evaluate whether the company's intended production processes are suitable.

END

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PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Tc 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 TI 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															

139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 69	173 Yb 70	175 Lu 71
227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

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